

Liuyin Yang

PhD Researcher · Brain Foundation Models & Brain–Computer Interfaces · KU Leuven

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I build **brain foundation models** spanning ECoG, EEG, sEEG and MEG for brain–computer interfaces: transferring large models pretrained on scalp EEG to **intracranial recordings** (CORTEG, NeurIPS 2026), testing when foundation models beat traditional decoders (ICLR 2026), and decoding auditory and motor signals in real-time BCI systems. Winner of the **NeurIPS 2025 EEG Foundation Challenge**.

Education

2021 – now — **PhD, Biomedical Sciences (Medicine)** · KU Leuven, Belgium
EEG-based brain–computer interfaces. Advisor: Prof. Marc M. Van Hulle.

2020 – 2021 — **Advanced Master of Artificial Intelligence** · KU Leuven
Big Data Analytics · Cum laude (75.33%)

2019 – 2020 — **MSc, Electronics & ICT Engineering Technology** · KU Leuven
Cum laude (73.85%)

2017 – 2019 — **BSc, Electronics Engineering** · KU Leuven
Cum laude (72.58%)

2015 – 2017 — **BEng, Electrical Engineering & Automation** · Chongqing University, China
GPA 3.3 / 4

Research Experience

2021 – now — **PhD Researcher** · Computational Neuroscience Group, KU Leuven

- Cross-modality transfer of scalp-EEG foundation models to intracranial (ECoG) recordings with an electrode-aware spatial adapter and dual-stream tokenizer (CORTEG, NeurIPS 2026).
- Pretrained a large spatiotemporal transformer on EEG with masked autoencoding; scaled pretraining on the VSC Tier-2 HPC; benchmarked EEG foundation models against traditional decoders across BCI tasks (ICLR 2026).
- 1st place, NeurIPS 2025 EEG Foundation Challenge (cross-task and cross-subject EEG decoding).
- WaveNet-based auditory EEG regression model, 2nd place in the ICASSP 2023 Auditory EEG Grand Challenge.
- Online motor-imagery BCI for real-time navigation in Google Street View; binocularly incongruent SSVEP BCI in VR (Meta Quest Pro).

2019 – 2020 — **Master's Thesis Researcher** · KU Leuven, Engineering Technology
Real-time adaptive-filter speech enhancement (noise reduction) on a DSP board, deployed in an IoT communication system.

Grants & Fellowships

2025 – 2026 — **VSC Compute Projects** · Flemish Supercomputer Center (VSC); 72,119 GPU hours in total
2026_064: 39,961 h · 2025_038: 24,661 h · 2025_500: 7,497 h (special grant, NeurIPS 2025 challenge)

2021 – 2025 — **FWO PhD Fellowship Strategic Basic Research (SB)** · Research Foundation – Flanders (FWO)

Awards & Honors

2025 — **1st Place — EEG Foundation Challenge 2025** · NeurIPS 2025 Competition Track (From Cross-Task to Cross-Subject EEG Decoding)

2023 — **2nd Place — ICASSP 2023 Grand Challenge** · Auditory EEG Decoding Challenge

2019 – 2021 — **Cum laude in three consecutive KU Leuven degrees** · Advanced Master AI, MSc Electronics & ICT, BSc Electronics

Publications

CONFERENCE PAPERS

NeurIPS 2026 — **CORTEG: Foundation Models Enable Cross-Modality Representation Transfer from Scalp to Intracranial Brain Recordings.** L Yang*, Q Sun*, B Van Dyck, EC Merino, MM Van Hulle (*equal contribution).
Conference on Neural Information Processing Systems (NeurIPS) 2026, poster.

ICLR 2026 **Are EEG Foundation Models Worth It? Comparative Evaluation with Traditional Decoders in Diverse BCI Tasks.** L Yang, Q Sun, A Li, MM Van Hulle. *International Conference on Learning Representations (ICLR) 2026,* poster.

ICORR 2025 **On the Impact of Proprioception in EEG Representations and Decoding during Human-Hand Exoskeleton Interaction.** Q Sun, EC Merino, L Yang, A Faes, MM Van Hulle. *International Conference on Rehabilitation Robotics (ICORR) 2025.*

ICASSP 2023 **Decoding Auditory EEG Responses Using an Adapted WaveNet.** B Van Dyck, L Yang, MM Van Hulle. *IEEE ICASSP 2023, Auditory EEG Grand Challenge (*2nd place*).*

JOURNAL ARTICLES

2024 **Unraveling EEG Correlates of Unimanual Finger Movements: Insights from Non-Repetitive Flexion and Extension Tasks.** Q Sun, EC Merino, L Yang, MM Van Hulle. *Journal of NeuroEngineering and Rehabilitation, 2024.*

2024 **Binocularly Incongruent, Multifrequency-Coded SSVEP in VR: Feasibility and Characteristics.** L Yang, Q Sun, MM Van Hulle. *Journal of Neural Engineering, 2024.*

2024 **Sea-Wave: Speech Envelope Reconstruction from Auditory EEG with an Adapted WaveNet.** L Yang, B Van Dyck, MM Van Hulle. *IEEE Open Journal of Signal Processing, 2024.*

2023 **Real-Time Navigation in Google Street View Using a Motor Imagery-Based BCI.** L Yang, MM Van Hulle. *Sensors, 2023.*

2021 **Chronic Study on Brainwave Authentication in a Real-Life Setting: An LSTM-Based Bagging Approach.** L Yang, A Libert, MM Van Hulle. *Biosensors, 2021.*

WORKSHOP PAPERS

ICLR 2026 W **Do We Need Domain-Specific Time-Series Models? Insights from EEG Classification Benchmarks.** L Yang, Q Sun, MM Van Hulle. *1st ICLR Workshop on Time Series in the Age of Large Models (TSALM) 2026,* poster.

ICLR 2026 W **Learning Robust EEG Representations with a Large Spatiotemporal Transformer as a Foundation Model.** L Yang, MM Van Hulle. *1st ICLR Workshop on Time Series in the Age of Large Models (TSALM) 2026.*

ICLR 2026 W **Large EEG Foundation Model Learns Informative Low-Frequency Representations from Intracranial Brain Signals.** Q Sun, L Yang, EC Merino, MM Van Hulle. *1st ICLR Workshop on Time Series in the Age of Large Models (TSALM) 2026,* poster.

PREPRINTS & DATASETS

2026 **Characterization of Speech Imagery in Scalp EEG and Comparison with Motor Imagery.** B Van Dyck, L Yang, Q Sun, A Li, MM Van Hulle. *arXiv preprint* arXiv:2606.12223.

2025 **CoMET: A Contrastive-Masked Brain Foundation Model for Universal EEG Representation.** A Li, Z Wang, L Yang, Z Wang, T Xu, H Hu, MM Van Hulle. *arXiv preprint* arXiv:2509.00314.

2026 **A Multi-Paradigm EEG Dataset for Multi-Task BCI Applications.** L Yang, B Van Dyck, Q Sun, A Li, MM Van Hulle. *KU Leuven Research Data Repository,* doi:10.48804/3EUSTS.

Talks, Teaching & Service

Dec 2025 **Spotlight talk — EEG Foundation Challenge 2025** · NeurIPS 2025 Competition Track, San Diego

2025 **EEG Foundation Models** · IEEE NER 2025 Workshop

2025 **Scaling EEG Foundation Model Pretraining** · KU Leuven / UHasselt Tier-2 HPC User Day

recurring **Teaching Assistant** · Brain–Computer Interfaces (MSc course), KU Leuven

2023 – 2025 **Master's thesis daily supervisor** · VR imagined-speech speller and brain connectivity with a brain foundation model (2024–25); VR-based binocular SSVEP system (2023–24)

reviewing **Reviewer** · journals: JNE, IEEE TPAMI · conferences: ICLR 2027, ICML 2026 Workshop FMSD

Skills

Deep learning PyTorch, distributed training (DDP), transformers, self-supervised learning

EEG & lab EEG experiment design and data collection, signal processing, Unity (VR — Meta Quest Pro), Android Studio

Programming Python, MATLAB, Octave, Java, C / C++ / C#, Assembler

Hardware Vivado (FPGA), DSP board (C5515 eZdsp), Arduino, Eagle (PCB), microcontrollers

Languages English — proficient · Chinese — proficient · French — intermediate (B1)